19. Theory of Games

The interdependence of firms in oligopolistic markets and the inherent uncertainty about competitors' reactions to any course of action adopted by a firm cannot be analysed effectively by the traditional tools of economic theory.

Economists have developed collusive models, limit-pricing models, managerial models, and behavioural models, but these do not provide a general theory of oligopoly in the sense that none of these models could fully explain the decision-making process of oligopolists.

A different approach to the study of the oligopoly problem is provided by the theory of games. The first systematic attempt in this field is von Neumann's and Morgenstern's *Theory of Games and Economic Behaviour*, published in 1944. Since that time numerous economists have developed models of oligopolistic behaviour based on the theory of games. Perhaps the most prominent proponent of the games-theory approach is Martin Shubik,[1] who seems to believe that the only hope for the development of a general theory of oligopoly is the games theory. Despite such enthusiasts the games-theory approach has not yet provided results which could lead to a general theory of oligopoly. The main contribution of this approach is that it can use high-speed computers to conduct experiments of oligopolistic behaviour. Such controlled experiments cannot be conducted in the real business world. Thus the computerised study of oligopolistic behaviour is an extremely useful tool which might lead to generalisations about the decision-making process of oligopolists in real world situations, and thus provide the basis for a theory of oligopoly. This development, however, is still in its infancy, and has not as yet gained much acclaim by the majority of economists.

In this chapter we provide an elementary exposition of the simplest models of games theory.

I. SOME DEFINITIONS

The firm has various *instruments* or *policy variables* with which it can pursue its goals. The most important are the price, quantity and style of its products, advertising and other selling activities, research and development expenditures, channels for selling the product(s), and changes in the number of products (discontinuation of an old product or introduction of new ones).

A *strategy* is a specific course of action with clearly defined values for the policy variables. For example, a strategy may consist of setting a price of £3·95, spending £2000 on advertising, making a change in the packaging of the product, and selling it in

[1] M. Shubik, *Strategy and Market Structure* (John Wiley, 1959). Also M. Shubik, 'A Curmudgeon's Guide to Microeconomics', *Journal of Economic Literature* (1971).

supermarkets. Another strategy may involve leaving the price unchanged, spending £1000 on advertising, devoting £2000 to research on a new product, and so on. To each of these strategies a competitor may react in different ways, that is, by adopting different strategies. He may decide to adopt the same or a different course of action than the one adopted by the other firm. Thus each firm has various strategies open to it, and in any particular case it will adopt the one that will seem most advantageous under the circumstances.

The *payoff* of a strategy is the 'net gain' it will bring to the firm for any given counter-strategy of the competitor(s). This gain is measured in terms of the goal(s) of the firm. For example, if the goal of the firm is to maximise its profit, the payoffs of a strategy will be measured in terms of profit levels that it yields; if the goal is maximisation of the market share, the payoffs will be measured as the actual shares that the strategy will secure to the firm adopting it.

The *payoff matrix* of a firm is a table showing the payoffs accruing to this firm as a result of each possible combination of strategies adopted by it and by its rival(s). For example, assume that there are two firms in the industry. Firm I has to choose among five strategies ($A_1, A_2, \ldots, A_5$) and Firm II can react by adopting any one of six strategies open to it ($B_1, B_2, \ldots, B_6$). Thus to each strategy of Firm I there are six possible counter-strategies of Firm II, and similarly to each strategy of Firm II there are five counter-strategies of the rival Firm I. Thus the payoff matrix of each firm will include $5 \times 6 = 30$ payoffs, corresponding to the results of each possible combination of strategies selected by both rivals.

Let us denote each payoff G_{ij}, where i refers to the strategy adopted by Firm I and j to the counter-strategy adopted by Firm II. Thus for the above example the payoff matrix for Firm I will be of the general form of table 19.1. If Firm I adopts strategy A_1 and its rival reacts by adopting among the strategies open to it B_5, the payoff (gain) of Firm I will be G_{15}. If Firm I chooses strategy A_4 and its rival reacts with strategy B_6, the payoff of Firm I will be G_{46}, and so on.

In the theory of games the firms in oligopolistic markets are treated as players in a chess game: to each movement by one player the other may choose among several counter-movements. The counter-movements of rivals are probable but not certain.

Table 19.1 Payoff matrix of Firm I

Firm I's strategies \ Firm II's strategies	B_1	B_2	B_3	B_4	B_5	B_6
A_1	G_{11}	G_{12}	G_{13}	G_{14}	G_{15}	G_{16}
A_2	G_{21}	G_{22}	G_{23}	G_{24}	G_{25}	G_{26}
A_3	G_{31}	G_{32}	G_{33}	G_{34}	G_{35}	G_{36}
A_4	G_{41}	G_{42}	G_{43}	G_{44}	G_{45}	G_{46}
A_5	G_{51}	G_{52}	G_{53}	G_{54}	G_{55}	G_{56}

Yet it is possible to choose a strategy which (under certain conditions) will maximise the firm's expected 'gain', after making due allowance for the effects of rivals' probable reactions.

II. TWO-PERSON ZERO-SUM GAME

A. CERTAINTY MODEL

The simplest model is a duopoly market in which each duopolist attempts to maximise his market share. Given this goal, whatever a firm gains (by increasing its share of the market) the other firm loses (because of the decrease in its share). Thus any gain of one rival is offset by the loss of the other, and the net gain sums up to zero. Hence the name 'zero-sum game'.

The assumptions of the model are:

1. The firms have a given, well-defined goal. In our particular example the goal is maximisation of the market share.
2. Each firm knows the strategies open to it and to its rival, or concentrates on the most important of these strategies.
3. Each firm knows with certainty the payoffs of all combinations of the strategies being considered. This implies that the firm knows its total revenue, total costs and total profit from each combination of strategies.
4. The actions chosen by the duopolists do not affect the total size of the market.
5. Each firm chooses its strategy 'expecting the worst from its rival', that is, each firm acts in the most conservative way, expecting that the rival will choose the best possible counter-strategy open to him. This behaviour is defined as 'rational'.
6. In the zero-sum game there is no incentive for collusion, given assumption 4, since the goals of the firms are diametrically opposed.

In order to find the equilibrium solution we need information on the payoff matrix of the two firms. In our example the payoffs will be shares of the market resulting from the adoption of any two strategies by the rivals. Assume that Firm I has four strategies open to it and Firm II has five strategies. The payoff matrices of the duopolists are shown in tables 19.2 and 19.3.

Table 19.2 Payoff matrix of Firm I

Firm I's strategies	Firm II's strategies				
	B_1	B_2	B_3	B_4	B_5
A_1	(0·10)	0·20	0·15	0·30	0·25
A_2	0·40	(0·30)	0·50	0·55	0·45
A_3	0·35	0·25	(0·20)	0·40	0·50
A_4	0·25	(0·15)	0·35	0·60	0·20

Table 19.3 Payoff matrix of Firm II

Firm I's strategies	Firm II's strategies				
	B_1	B_2	B_3	B_4	B_5
A_1	0·90	0·80	0·85	0·70	0·75
A_2	**0·60**	**0·70**	**0·50**	0·45	0·55
A_3	0·65	0·75	0·80	0·60	**0·50**
A_4	0·75	0·85	0·65	**0·40**	0·80

Clearly the sum of the payoffs in corresponding cells of the two payoff tables add up to unity, since the numbers in these cells are shares, and the total market is shared between the two firms. In general, in the two-person zero-sum game we need not write both payoff matrices because of the nature of the game: the goals are opposing, and, in our example, the payoff table of Firm I contains indirectly information about the payoff of Firm II. Still we start by showing both tables, and then we show how the equilibrium solution can be found from only the first payoff matrix.

Choice of strategy by Firm I

Firm I examines the outcomes of each strategy open to it. That is, Firm I examines each row of its payoff matrix and finds the most favourable outcome of the corresponding strategy, because the firm expects the rival to adopt the most advantageous action open to him. This is the behavioural rule implied by assumption 5 of this model. Thus:

If Firm I adopts strategy A_1, the worst outcome that it may expect is a share of 0·10 (which will be realised if the rival Firm II adopts its most favourable strategy B_1).

If Firm I adopts strategy A_2, the worst outcome will be a share of 0·30 (if the rival adopts the best action for him, B_2).

If Firm I adopts strategy A_3, the worst outcome will be a share of 0·20 (if Firm II chooses the best open alternative, B_3).

If Firm I adopts strategy A_4, the worst outcome will be a share of 0·15 (which would be realised by action B_2 of Firm II).

Among all these *minima* (that is, among the above worst outcomes) Firm I chooses the *maximum*, the 'best of the worst'. This is called a *maximin strategy*, because the firm chooses the maximum among the minima. In our example the maximin strategy of Firm I is A_2, that is, the strategy which yields a share of 0·30.

Choice of strategy by Firm II

Firm II behaves in exactly the same way. The only difference is that Firm II examines the *columns* of its payoff table, because these columns include the results–payoffs of each of the strategies open to Firm II. For each strategy, that is, for each column, Firm II finds the worst outcome (on the assumption that the rival will choose the best),

Table 19.4 Combined payoff matrix

Firm I's strategies (maximin behaviour)	Firm II's strategies (minimax behaviour)				
	B_1	B_2	B_3	B_4	B_5
A_1	(0·10)	0·20	0·15	0·30	0·25
A_2	[0·40]	[(0·30)]	[0·50]	0·55	0·45
A_3	0·35	0·25	(0·20)	0·40	[0·50]
A_4	0·25	(0·15)	0·35	[0·60]	0·20

and among these worst outcomes Firm II chooses the best. Thus, if Firm II uses its own payoff table, its behaviour is a maximin behaviour identical to the behaviour of Firm I.

However, as we said earlier, in the zero-sum game only one payoff matrix is adequate for the equilibrium solution. In our example the first payoff table will be used not only by Firm I but also by Firm II. Thus concentrating on the first payoff table we may re-state the decision-making process of Firm II as follows. Firm II examines the columns of the (first) payoff matrix because these columns contain the information about the payoffs of its strategies. For each column-strategy Firm II finds the maximum payoff (of Firm I) because this is the worst situation the firm (II) will face if it adopts the strategy correponding to that column. Thus for strategy B_1 the worst outcome (for Firm II) is 0·40; for strategy B_2 the worst outcome is 0·30; for strategy B_3 the worst outcome is 0·50; for strategy B_4 the worst result is 0·60; for strategy B_5 the worst result is 0·50. Among these maxima of each column-strategy Firm II will choose the strategy with minimum value. Thus the strategy of Firm II is a *minimax* strategy, since it involves the choice of a minimum among the maxima payoffs. (Table 19.4.)

It should be stressed that although different terms are used for the choice of the two firms (maximin behaviour of Firm I, minimax behaviour of Firm II), the behavioural rule for both firms is the same: each firm expects the worst from its rival.

In our example the equilibrium solution is strategy A_2 for Firm I and B_2 for Firm II. This solution yields shares 0·30 for Firm I and 0·70 for Firm II. It is an equilibrium solution because it is the preferred one by both firms. This solution is called the '*saddle point*', and the preferred strategies A_2 and B_2 are called '*dominant strategies*'.

It should be clear that there exists no such equilibrium (saddle) solution if there is no payoff which is preferred by both firms simultaneously. Under certain mathematical conditions other solutions and strategy choices can be determined. The analysis of the resulting *mixed strategies* requires a sophisticated exposition of utility theory and random selection which is beyond the scope of this book.

B. UNCERTAINTY MODEL

The assumption that each firm knows with certainty the exact value of the payoff of each strategy is unrealistic. The most probable situation in the real business world is

that the firm, by adopting a certain strategy, may expect a range of results for each counter-strategy of the rival, each result with an associated probability. Thus the payoff matrix is constructed so as to include the *expected value* of each payoff. The expected value is the sum of the products of the possible outcomes of a pair of strategies (adopted by the two firms) each multiplied by its probability:

$$E(G_{ij}) = g_{1i}P_1 + g_{2i}P_2 + \cdots + g_{ni}P_n$$

$$= \sum_{s=1}^{n} g_{si}P_s$$

where g_{si} = the sth of the n possible outcomes of strategy i of Firm I (given that Firm II has chosen strategy j)
P_s = the probability of the sth outcome of strategy i

For example, assume that Firm I chooses strategy A_1 and Firm II reacts with strategy B_1. This pair of simultaneous strategies may yield the shares for Firm I each with a certain probability, shown in the second column of table 19.5. Thus the expected payoff of the pair of strategies A_1 and B_1 is

$$E(G_{11}) = (0{\cdot}00)(0{\cdot}00) + (0{\cdot}05)(0{\cdot}05) + (0{\cdot}15)(0{\cdot}05) + \cdots + (0{\cdot}95)(0{\cdot}02) + (1)(0)$$
$$= 0{\cdot}458$$

Table 19.5

Possible shares of Firm I for the pair of strategies A_1, B_1	*Probability of each share*
0·00	0·00
0·05	0·05
0·15	0·05
0·25	0·10
0·35	0·15
0·45	0·25
0·55	0·20
0·65	0·10
0·75	0·05
0·85	0·03
0·95	0·02
1·00	0·00
	$\Sigma P_i = 1$

In a similar way we find the expected payoff of all combinations of strategies. Given the matrix of expected payoffs, the behavioural pattern of the firms is the same as in the certainty model.
That is:

Firm I adopts the *maximin strategy*. It finds for *each row* the minimum expected payoff, and among these minima the firm chooses the one with the highest value (the maximum among the minima).

Firm II adopts the *minimax strategy*. It finds for *each column* the maximum expected payoff, and among these maxima Firm II chooses the one with the smallest value (the minimum among the maxima).

Although the uncertainty zero-sum game seems simple, its assumptions are quite stringent:

1. The firms maximise their expected payoffs.
2. The zero-sum game assumes that both firms assign the same probability to each pair of payoffs; they make the same judgement. This implies that the firms must have the same information and the same objective criteria with which to evaluate the probabilities of the different payoffs. Otherwise the probability distribution of the payoffs will not be objective.
3. The firms maximise their total utility, and the utility of each payoff is proportional to the value assumed by the payoff.

The above assumptions are clearly strong and unrealistic. Furthermore, the basic condition of the zero-sum game, that the 'gain' of one firm is equal to the 'loss' of the other, is rarely met in the real business world. Usually the 'gains' are not 'offset' by equal 'losses'. Only in the case of a share goal, and in the rare case of extinction tactics, do we have a zero-sum game. In most cases we have a non-zero-sum game.

III. NON-ZERO-SUM GAME

We will illustrate this model with a duopolistic market in which the firms aim at the maximisation of their profit. Their products are close substitutes so that if their prices differ the firm with the lower price will supply the largest part of the market. It is assumed that the firms will use price as their instrumental variable. For simplicity we assume that each firm can charge two prices (either £3 or £5), that is, there are two strategies open to each competitor. Each firm has a different cost structure and the market size is affected by the rivals' combined action. Under these conditions the payoff matrix of each firm is expressed in terms of levels of profit, and the gains of one rival need not be (and in our example are not) equal to the losses of the other. The payoff matrices of the two firms are shown below in tables 19.6 and 19.7, and are subsequently combined in a single table. (Table 19.8.)

Table 19.6 Firm I's payoff matrix
(level of profits of I)

Firm I's strategies	Firm II's strategies		
		$P = 5$	$P = 3$
	$P = 5$	$\Pi_A = 90$	$\Pi_A = 50$
	$P = 3$	$\Pi_A = 150$	$\Pi_A = 80$

Table 19.7 Firm II's payoff matrix
(levels of profit of II)

Firm I's strategies	Firm II's strategies		
		$P_B = 5$	$P_B = 3$
	$P_A = 5$	$\Pi_B = 110$	$\Pi_B = 120$
	$P_A = 3$	$\Pi_B = 60$	$\Pi_B = 100$

Table 19.8 Combined payoff matrix

Firm I's strategies	Firm II's strategies: $P_B = 5$	Firm II's strategies: $P_B = 3$
$P_A = 5$	$\Pi_A = 90$ ¦ $\Pi_B = 110$ Joint $\Pi = 200$	$\Pi_A = 50$ ¦ $\Pi_B = 120$ Joint $\Pi = 170$
$P_A = 3$	$\Pi_A = 150$ ¦ $\Pi_B = 60$ Joint $\Pi = 210$	$\Pi_A = 80$ ¦ $\Pi_B = 100$ Joint $\Pi = 180$

The behavioural rule is the same for both firms: each expects the worst from the rival. The choice of Firm I is a *maximin strategy*. If Firm I sets the price of £5 the minimum gain is £50; if it sets $P = 3$ its minimum profit is £80. Among these two minima the firm chooses the maximum, that is, the preferred strategy by Firm I is $P = 3$.

The choice of Firm II is also a *maximin strategy*. If Firm II charges a price of £5 the worst it can expect is a profit of £60; if it charges a price of £3 the minimum level of profit is £100. Among these minima the firm will choose the maximum, that is, Firm II will choose the price of £3.

Under these circumstances there is a unique equilibrium price (£3) which will be adopted by both firms. Thus the strategy $P =$ £3 is a *dominant strategy*. Yet with this strategy both firms are in a worse situation as compared to the alternative strategy $P = 5$, since both realise a lower profit. And of course the industry (joint) profit is not maximised. The conservative maximin strategy is not the optimal solution in this case. If the firms colluded and both charged the higher price of £5 the joint profit and their individual profits would be higher ($90 > 80$, $110 > 100$, and $200 > 180$). Thus while the maximin strategy provides an optimal solution in the zero-sum game, this may not be so in the variable-sum game.

Many other oligopolistic actions may be analysed with the above apparatus of the theory of games. For example, advertising campaigns, change in style or diversification, research and development expenditures may be examined by the principles of the games theory. In most real-world cases we see that firms choose strategies which do not maximise their profits: advertising or new-product rivalry often lead to excessive increases in costs of all firms in the industry. Such situations may be explained by the conservative behaviour of maximin strategies.

In many oligopolistic situations firms seem to avoid the unfavourable outcomes predicted by the maximin–minimax behaviour of the theory of games. Several reasons have been offered for these cases. Firstly, the duration of rivalry. If rivalry has been continuous for a considerable time-period the rivals 'learn' to predict the reactions of each other and this leads to the avoidance of moves which, from well-established past patterns, have proved disadvantageous to all parties. Secondly, the stability of tastes and processes. Firms are more likely to avoid mutually damaging actions in a market where demand does not change and technological progress is slow. On the other hand, in markets with frequent changes of tastes it is almost certain that firms will adopt maximin strategies despite their mutually unfavourable results. Thirdly, the existence of common sources of information and communication between rivals. If the rivals lack information it is most natural to fear the worst (maximin assumption) from the competitors and reach suboptimal solutions. Fourthly, time-lags are important in deciding which strategy to adopt. If the imitation of an action (for example, a new product, a new process) is easy the firms will recognise that they have little to gain and much to lose by aggressive action and hence will adopt maximin strategies, which are by their nature

conservative. If, however, an action cannot easily or quickly be imitated the firms will tend to abandon maximin attitudes and adopt actions which lead to more favourable positions for themselves, instead of expecting the worst from their rivals, for the simple reason that the rivals cannot quickly adopt the most advantageous action.

In summary: games theory has not provided a general theory of oligopolistic behaviour. However, the games-theory approach has been able to explain some real-world situations. It has helped in showing that there are strong incentives for collusion in several oligopolistic situations. Perhaps the most important contribution of this theory is that it has led to controlled experiments in the study of firms' behaviour. By expressing the alternatives open to a firm and its rivals in the form of payoff matrices it has made possible the examination of an increasing number of alternatives with controlled experiments based on the use of high-speed computers. Playing 'monopoly games' with computers has been a fashionable tendency in many American universities. Such 'controlled' studies have provided insight into various aspects of the decision-making process of individuals in uncertain situations similar to those facing the firms in the real world. Yet games theory is still a long way from providing a comprehensive theory of oligopolistic behaviour.

IV. THE PRISONER'S DILEMMA: A DIGRESSION

We saw that in most cases of variable-sum games the maximin–minimax behaviour leads the rivals to suboptimal solutions, that is, to situations worse than they need be. These examples and their suboptimal solutions are a special case of the general group of problems which are known as *Prisoner's Dilemma Games.* A brief exposition of the original prisoner's dilemma might help the understanding of the behaviour of firms faced by uncertainty about their rivals' action.

Two criminals are arrested after committing a big bank robbery. However, the evidence is not adequate to make the robbery charge stand unless one or both criminals confess. Each suspect is interrogated in isolation from his companion so that no communication is possible between them. The District Attorney promises no punishment for the suspect who confesses and a heavy sentence of, say, twenty years' imprisonment for the other party. If both suspects do not confess, both will go free. If both confess, they will get the sentence prescribed by the law for the crime of robbery, for example ten years' imprisonment. Thus each suspect has two 'strategies' open to himself, to confess or not to confess, and is faced with the dilemma: to confess (and go free if the other does not confess, or get the ten-year sentence) or not to confess (and go free if the other does not confess, or get the heavy twenty-year sentence if he is betrayed by the other suspect). The payoff matrix of the two prisoners is shown in table 19.9. Given the lack of communication between the suspects and the uncertainty as to the 'loyalty' of the other prisoner, each one of them prefers to adopt the second strategy, that is, to

Table 19.9

Prisoner *A*'s strategies	Prisoner *B*'s strategies: No confession	Prisoner *B*'s strategies: Confession
No confession	*A* 0, *B* 0	*A* 20, *B* 0
Confession	*A* 0, *B* 20	*A* 10, *B* 10

confess, so that both get a ten-year sentence. Clearly this is a worse situation as compared with the adoption of the 'no confession' strategy by both robbers. The 'dominant' strategy, which implies the rule 'expect the worst from the other(s)' (the maximin assumption) leads to a worse position than the robbers need be in! If communication were possible, or if from past experience the fellows had learned to trust each other, they would both plead 'not guilty' and would go free, thus maximising their 'gains'.

The drawing of analogies concerning the conditions of uncertainty facing the firms in oligopolistic markets is straightforward.